# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

# CHEMISTRY

Past Paper 2024 • Paper Code: 1HA-I 24005 — 30052 (Version B) / 32053 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

## VERSION B

## CHEMISTRY HSSC-I — SECTION – A (Objective Type)

**Time Allowed:** 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 30052 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. A liquid is thought to be pure ethanol. Which of the following is best way to test its purity?

(A) Dehydrating it with H2SO4
(B) Measuring its boiling point
(C) Reacting with ethanol
(D) Burning it completely

Q2. Volume occupied by 2 g of SO2 gas at S.T.P will be:

(A) 17 dm^3
(B) 0.7 dm^3
(C) 0.7 cm^3
(D) 700 dm^3

Q3. The largest/maximum number of molecules are present in:

(A) 180 g of C6H12O6
(B) 44 g of CO2
(C) 98 g of H2SO4
(D) 36 g of H2O

Q4. If ^29_61 Cu and ^79_197 Au are used as anodes for production of x-rays, then "Cu" will produce the x-rays of:

(A) High energy
(B) Shorter wavelength
(C) Longer wavelength
(D) High frequency

Q5. The third line of Balmer series appears due to the transition of electron from:

(A) n3 → n1
(B) n4 → n2
(C) n5 → n2
(D) n5 → n1

Q6. Which of the following covalent bond has lowest bond energy?

(A) H - I
(B) H - F
(C) H - Cl
(D) H - Br

Q7. Shape of PH3 molecule is:

(A) T-shape
(B) Octahedral
(C) Pyramidal
(D) Triangular planar

Q8. Which pair of gases diffuses with the same rate at same temperature and pressure?

(A) CO and PH3
(B) CO and NO
(C) NO and C2H6
(D) NH3 and PH3

Q9. Which of the following is NOT present in plasma?

(A) Neutral atoms
(B) Electrons
(C) Cations
(D) Anions

Q10. Which of the following processes is NOT endothermic?

(A) Vaporization
(B) Condensation
(C) Fusion
(D) Sublimation

Q11. If crystallization of "NaCl" is carried out in the presence of urea then its shape will be:

(A) Octahedral
(B) Cubic
(C) Tetrahedral
(D) Pyramidal

Q12. The strong conjugate base in the given is:

(A) CO3^-2
(B) NO3^-1
(C) Cl^-1
(D) SO4^-2

Q13. A reaction will proceed in forward direction to maximum extent if value of Kc is:

(A) 10^-2
(B) 10^-30
(C) 1
(D) 10^+30

Q14. For a reaction "A → Product", doubling the concentration of "A" quadruples the rate. The reaction is:

(A) Zero order
(B) 1st order
(C) 2nd order
(D) 3rd order

Q15. Swelling of dead bones in water is due to:

(A) Osmosis
(B) Evaporation
(C) Decomposition
(D) Boiling

Q16. Which of the following is NOT a state function?

(A) Work
(B) Temperature
(C) Pressure
(D) Volume

Q17. "Zn" acts as anode in Daniel cell but acts as cathode when coupled with Aluminum electrode. This is because:

(A) E° of Zn = 0
(B) E° of Zn > E° of Al
(C) E° of Zn < E° of Al
(D) E° of Zn = E° of Al

## CHEMISTRY HSSC-I — SECTIONS B & C (Subjective Type)

**Time Allowed:** 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 24005-(B) (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 42) — Q.2 Answers the following questions briefly. (14 x 3 = 42)

درج سوالات کے مختصر جوابات تحریر کریں۔

(i) [03] Calculate the volume of oxygen produced by decomposition of 3.04×10^24 formula units of KClO3 according to given equation. 2KClO3 → 2KCl + 3O2 — OR (03) — Calculate the wave number ($\bar{\nu}$) of H_α in Balmer series and second line in Paschen series of hydrogen spectrum.

(ii) [03] In an industrial process 40 g of "H2" produces 100 g of "NH3". Calculate the percentage yield of this reaction. N2 + 3H2 → 2NH3 — OR (03) — Justify the following statements with reference to azimuthal quantum number: (i) s-orbital has maximum two $e^-$ (ii) p-orbital can accommodate maximum six electrons

(iii) [03] Calculate the number of molecules of SO2 gas if its volume is 500 cm^3 at S.T.P. — OR (03) — Why CO2 is linear while H2O is bent or V-shape, although atomicity of both molecules is same?

(iv) [03] Calculate the number of molecules of CO2 when 4.8×10^24 molecules of CH4 reacts with excess of water according to following reaction. CH4 + 2H2O → CO2 + 4H2 — OR (03) — Draw the shapes of following molecules according to VSEPR theory: (i) SO2 (ii) H2S (iii) CBr4

(v) [03] Briefly describe the following: (i) Line spectrum (ii) Stark effect (iii) Continuous spectrum — OR (03) — Distinguish between 'Sigma' and 'Pi bond' in three ways.

(vi) [03] Calculate the average molar mass of air at sea level at 0°C, if density of air is 1.29 kg/m^3. — OR (03) — Justify the following statements: (i) Petrol evaporates earlier than water (ii) Water has low vapour pressure than ethyl alcohol

(vii) [03] What is the effect on the volume of gas if you simultaneously: (i) Its pressure is halved and its kelvin temperature is doubled (ii) Its pressure is doubled and its kelvin temperature is doubled — OR (03) — Why heat of vaporization (ΔH_v) is always greater than heat of fusion (ΔH_f) for a substance?

(viii) [03] Calculate the numerical value of general gas constant "R" for one mole of gas at S.T.P: (i) In SI units (ii) Pressure in atm, volume in dm^3 — OR (03) — Write any three characteristics of Plasma.

(ix) [03] Predict the shape of ZnS by using formula of radius ratio, if radius of Zn^+2 is 74 pm and radius of S^-2 is 184 pm. — OR (03) — If initial concentration of N2O4 in moles is "a" and "x" moles of it converted to NO2, then derive the general relation of equilibrium constant (Kc = 4x^2/v(a-x)) for following reaction: N2O4 ⇌ 2NO2

(x) [03] Calculate the value of K_p at 1050°C if K_c is 2.3×10^22 for following reaction. 2CO_(g) + O2_(g) ⇌ 2CO2_(g) — OR (03) — Write K_sp expressions for following compounds: (I) Ca3(PO4)2 (II) Na2SO4

(xi) [03] What is levelling effect of water? How this effect is compensated? — OR (03) — Draw potential energy diagram for both exothermic and endothermic reactions according to collision theory.

(xii) [03] Rate equation for given reaction is R = K[NO]^2[H2]. Reaction occurs in two steps and oxygen atom is intermediate then write reaction mechanism. 2NO + 2H2 → N2 + 2H2O — OR (03) — How relative lowering of vapour pressure (ΔP/P° = X2) can be used to calculate molar mass of solute?

(xiii) [03] Describe phenol water system and explain upper consulate temperature. — OR (03) — Write thermochemical equations from the given temperature: (i) Standard enthalpy of formation of CaCO3 is -1207 kJ/mol (ii) Standard enthalpy of combustion of CH3COOH is -875 kJ/mol

(xiv) [03] Calculate E°_cell for Li - Zn cell and write cell reactions. E°_Li is -3.05 V and E°_Zn is -0.76 V. — OR (03) — Write chemical reactions that occur at cathode and anode in alkaline dry cell.

> Note: SECTION – C (Marks 26) — Attempt the following questions.

Q3. (Marks 1+3+3) Q.3 What is hybridization? Explain the hybridization of CH4 and BeCl2 with orbital diagrams. — OR (2+2+3) — Why real gases deviate from gas laws at low temperature and high pressure? Also explains the deviation by general graphical representation of compressibility factor versus pressure.

Q4. (Marks 2+4) Q.4 State first law of thermodynamics. Write its mathematical expression with reference to heat and work. Explain it for a gas confined to a cylinder having a moveable piston, and derive the formula of W = PΔV for this system. — OR (3+3) — Balance the given redox equations by oxidation number method: (i) HNO3 + H2S → NO + S + H2O (ii) P + H2O + HNO3 → H3PO4 + NO

Q5. (Marks 1+2+3) Q.5 What is buffer solution? Write its types with composition. Explain buffer action when small amount of base is added in it. — OR (1+3+2) — What are colligative properties of solutions? Explain quantitative aspects of freezing point depression with general graphical representation to derive the molar mass of solute.

Q6. (Marks 3+4) Q.6 What are London dispersion forces? Explain any three factors which affect these forces with suitable example in each factor. — OR (2+2+3) — Differentiate between cubic close packing and hexagonal close packing in metals. Also compare metallic solids with molecular solids in three ways.

# VERSION D

## CHEMISTRY HSSC-I — SECTION – A (Objective Type)

**Time Allowed:** 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 32053 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. A device that converts the energy of combustion of Fuels like hydrogen directly into electrical energy is known as:

(A) Lead Storage cell
(B) Fuel cell
(C) NICAD cell
(D) Dry cell

Q2. Which of the following gases resembles ideal gas?

(A) HCl
(B) C4H10
(C) H2
(D) NH3

Q3. A beaker contains 9 g of water. The number of hydrogen atoms in it will be:

(A) 6.02×10^24
(B) 3.01×10^24
(C) 6.02×10^23
(D) 3.01×10^23

Q4. Which one of the following pairs of gases contains the same number of molecules at S.T.P?

(A) 28 g of N2 and 22 g of CO2
(B) 32 g of O2 and 32 g of N2
(C) 16 g of O2 and 14 g of N2
(D) 8 g of O2 and 22 g of CO2

Q5. According to Aufbau Principle a new electrons enters the orbital when:

(A) m + s is maximum
(B) m + s is minimum
(C) n + l is maximum
(D) n + l is minimum

Q6. The frequency of green light is 6×10^14 Hz. Its wavelength is:

(A) 5000 nm
(B) 50000 nm
(C) 5 nm
(D) 500 nm

Q7. In which of the following molecules central atom has sp^2 hybridization?

(A) BF3
(B) NH3
(C) H2O
(D) NF3

Q8. Which one of the following molecules has zero dipole moment?

(A) H2S (Bent)
(B) BF3 (Triangular Planar)
(C) NH3 (Pyramidal)
(D) H2O (Bent)

Q9. Which pair of the gases diffuses with the same rate at S.T.P?

(A) CO and PH3
(B) NH3 and PH3
(C) CO and NO
(D) NO and C2H6

Q10. To which of the following gaseous mixture, Dalton's law is NOT applicable?

(A) N2 and CO2
(B) NH3 and HCl
(C) H2 and N2
(D) Ne and He

Q11. Which of the following is NOT the unit of viscosity?

(A) poise
(B) Nm^-2s
(C) poise^-1
(D) kgm^-1s^-1

Q12. Which of the following pairs are isomorphs of each other?

(A) MgO and MgCl2
(B) NaCl and MgO
(C) MgO and CaCO3
(D) KNO3 and NaCl

Q13. Which of the following pairs constitute a buffer solution?

(A) NaCl + HCl
(B) Na2CO3 + H2CO3
(C) NaOH + HCl
(D) HNO3 + NH4NO3

Q14. The pH of a solution of HCl is 4. Molarity of solution will be:

(A) 0.04
(B) 4.0
(C) 0.4
(D) 0.0001

Q15. The rate of reaction _____ as the reaction proceeds.

(A) Remains the same
(B) Becomes infinite
(C) Increases
(D) Decreases

Q16. Which of the following is NOT the colligative property of solution?

(A) Optical activity
(B) Osmosis
(C) Elevation of Boiling point
(D) Depression of Freezing point

Q17. Born Haber cycle is the application of:

(A) Henry's law
(B) Moseley's law
(C) Faraday's law
(D) Hess's law

## CHEMISTRY HSSC-I — SECTIONS B & C (Subjective Type)

**Time Allowed:** 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 24005-(D) (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 42) — Q.2 Answers the following questions briefly. (14 x 3 = 42)

درج سوالات کے مختصر جوابات تحریر کریں۔

(i) [03] Calculate the mass of Hydrogen ions ($H^+$) produced by the ionization of 20 g of H2SO4. $H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{-2}$ — OR (03) — Calculate the radius of 3rd orbit for $_2He^{+1}$ r = $\varepsilon_o h^2 n^2 / z\pi m e^2$

(ii) [03] Calculate the number of formula units of MgS when 10 g of "Mg" reacts with 10 g of "S". Mg + S → MgS — OR (03) — How much "AgCl" will be formed by reacting 100 g of "AgNO3" (At.Wt, Ag = 107) with a solution of 50 g of "NaCl"? AgNO3 + NaCl → AgCl + NaNO3

(iii) [03] Calculate the volume of N2 gas for $3.01 \times 10^{23}$ molecules at S.T.P. — OR (03) — Justify the given order of energy of sub-shells according to n + l rule. (i) 3d > 4s (ii) 2p < 3s

(iv) [03] A photon of light has energy of $10^{-10}$ J. Convert this energy into frequency ($\nu$), wave length ($\lambda$) and wave numbers ($\bar{\nu}$) in Hz, meter and $m^{-1}$ respectively. — OR (03) — Justify the following statements: (i) Bond energy of H - H is greater than Cl - Cl (ii) Bond energy of H - Br is less than H - Cl

(v) [03] The dipole moment of HCl is 1.03 D and distance between atoms is 127 pm. Calculate the percentage ionic character of HCl bond. ($q = 1.6022\times10^{-19}$ C) (1 pm = $10^{-12}$ m) — OR (03) — Prove that kelvin temperature of a gas is the measurement of average kinetic energy of its molecules. (K.E ∝ T)

(vi) [1+2] What is the Charles law? Derive its critical form. (V_t = V_o/273 T) — OR (03) — Why a small droplet of water assumes nearly a spherical shape on the surface of a waxy bonnet of a car?

(vii) [03] Describe any two applications of Dalton's law of partial pressure. — OR (03) — Compare molecular and metallic solids in three ways.

(viii) [03] Write down the faulty postulates of Kinetic molecular theory. — OR (03) — Differentiate between Homogeneous and Heterogeneous equilibrium.

(ix) [03] Why boiling point of SiH4 is greater than CH4? Although both molecules are non-polar and have same atomicity. — OR (03) — Write K_sp expressions for following compounds: (i) Al(OH)3 (ii) Mg3(PO4)2

(x) [03] Differentiate liquid crystals from pure liquids and crystalline solids. (Any three differences) — OR (03) — Q' is called ion product. How is it helpful to determine the precipitation in a reaction by comparing it with K_sp?

(xi) [03] Write K_c expressions for following reactions and derive its unit: (i) C_(s) + H2O_(g) ⇌ CO_(g) + H2_(g) (ii) 3Fe_(s) + 4H2O_(g) ⇌ Fe3O4_(s) + 4H2_(g) — OR (03) — Derive the given relationship K_a × K_b = K_w for a conjugate acid base pair.

(xii) [03] What is meant by the following terms: (i) Order of reaction (ii) Initial rate of reaction (iii) Average rate of reaction — OR (1+2) — What is meant by the solvation? Briefly explain this term for ionic compounds.

(xiii) [03] Calculate the mass (w/w) percent of a solution containing 80 g sugar (C12H22O11) in 250 g of water. — OR (03) — Write thermochemical equations from the given information: (i) standard enthalpy of formation of Fe2O4 is -824 kJ/mol (ii) standard enthalpy of combustion of CH3COOH is -875 kJ/mol

(xiv) [03] Balance the following half reactions that take place in acidic medium: (i) $NO3^{-1}$ → NO2 (ii) $ClO4^{-1}$ → $ClO3^{-1}$ — OR (03) — Explain dry cell with the help of chemical reactions that occur at cathode and anode.

> Note: SECTION – C (Marks 26) — Attempt the following questions.

Q3. (Marks 1+3+3) Q.3 What is hybridization? Explain the hybridization of NH3 and BF3 with orbital diagrams. — OR (1+3+3) — State Dalton's law of partial pressure. Derive the relationship for partial pressure and number of moles for two supposed gases A and B.

Q4. (Marks 06) Q.4 Explain construction and working of lead storage battery with reactions that occur at anode and cathode during charging and discharging. — OR (2+2+2) — Predict the nature of the given salts (Acidic, Basic or Neural) and write

chemical equation when they are Hydrolysed in water: (i) $NH_4NO_3$ (ii) $MgSO_4$ (iii) $Na_2CO_3$

Q5. (Marks 1+3+2) Q.5 Why boiling point elevate when non-volatile solute is added in a solvent? Explain its quantitative aspects and derive the relationship for molar mass of solute using elevation of boiling point. — OR (3+3) — What is Hess's law? Give its mathematical expression and draw energy cycle for the given reaction: $C + O_2 \rightarrow CO_2$ $\Delta H° = -393.5$ kJ. Reaction can be carried out in two steps (i) $C + 1/2\ O_2 \rightarrow CO$ $\Delta H1° = -110.52$ kJ (ii) $CO + 1/2\ O_2 \rightarrow CO_2$ $\Delta H2° = -282.98$ kJ

Q6. (Marks 1+2+2+2) Q.6 What is meant by hydrogen bonding? How it explains the following: (i) Boiling point of water is higher than HF. (ii) Cleansing action of soap (iii) Solubility of covalent compounds — OR (1+3+3) — What is unit cell? Calculate the number of $Na^+$ and $Cl^-$ in one unit cell of Sodium Chloride.